BIKE STATION ANALYSIS

1. Project Overview and Objective

The **Bike Station Analysis** project focuses on analyzing bike availability, bike stand utilization, and city-wise station performance across multiple countries. The objective is to transform raw bike station data into meaningful insights using **data modelling , DAX measures, and interactive Power BI visualizations**.

2. Data Sources

Primary Dataset: Bike Station Sharing Dataset (Excel)

Data Type: Structured tabular data

Granularity: City-level and station-level

Key Attributes: Country, City, Bike availability, Bike stand capacity, Status, Date /Month

3. Problem Statement

Bike-sharing systems require efficient monitoring of bike availability and station utilization to ensure smooth operations. However, raw operational data alone does not provide actionable insights.

4. Attribute (Column /Features) Details:

Column Name	Description
Country	Country where the bike station is located
City	City name
Station Name	Name of the bike station
Total Bike Stands	Total capacity of the station
Available Bikes	Number of bikes currently available
Available Bike Stands	Free bike stands
Utilization %	Percentage of used bike stands
Status	Operational status of the station

5.Tools & Technologies

- **Power BI**– Data cleaning and initial preprocessing.
- **Power BI Desktop** – Data modeling, DAX calculations, and visualization.
- **DAX (Data Analysis Expressions)** – Measures and KPIs.

6.Data Pre-Processing (Power Query)

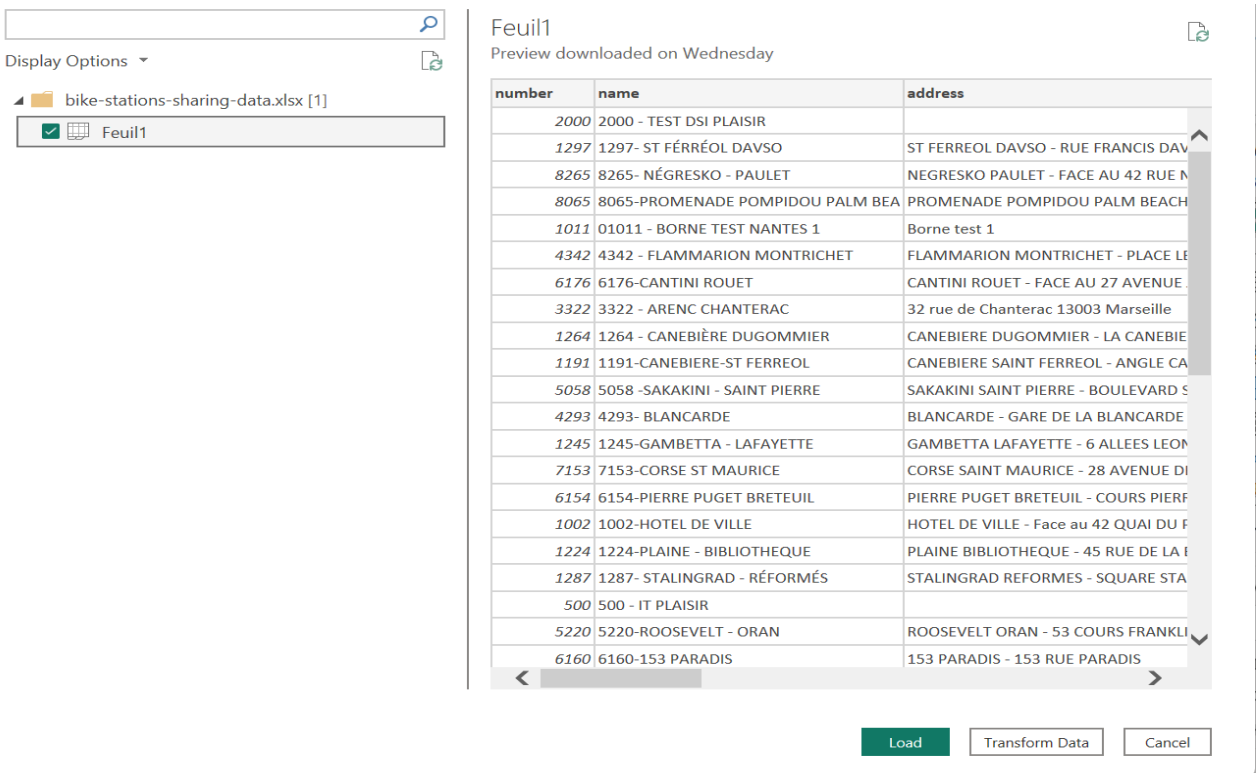
Task performed:

Data Cleaning

Removed duplicate records, Handled missing and null values, Standardized country and city names.

Data Transformation- Created calculated columns for utilization percentage , Filtered inactive or irrelevant station records.

Power BI Import



Using Custom Column to clean the name column by using

Text.Select([name],{“ “,”a”..”z”,”A”..”Z”})for text value

```
= Table.RenameColumns(#"Added Custom",{{"customer name", "Station  name"}})
```

	name	Station name
2000	2000 - TEST DSI PLAISIR	TEST DSI PLAISIR
1297	1297- ST FÉRRÉOL DAVSO	ST FRROL DAVSO
8265	8265- NÉGRESKO - PAULET	NGRESKO PAULET
8065	8065-PROMENADE POMPIDOU PALM ...	PROMENADE POMPIDOU PALM B...
1011	01011 - BORNE TEST NANTES 1	BORNE TEST NANTES
4342	4342 - FLAMMARION MONTRICHET	FLAMMARION MONTRICHET
6176	6176-CANTINI ROUET	CANTINI ROUET
3322	3322 - ARENC CHANTERAC	ARENC CHANTERAC
1264	1264 - CANEBIÈRE DUGOMMIER	CANEBIRE DUGOMMIER
1191	1191-CANEBIERE-ST FERREOL	CANEBIEREST FERREOL
5058	5058 -SAKAKINI - SAINT PIERRE	SAKAKINI SAINT PIERRE
4293	4293- BLANCARDE	BLANCARDE
1245	1245-GAMBETTA - LAFAYETTE	GAMBETTA LAFAYETTE
7153	7153-CORSE ST MAURICE	CORSE ST MAURICE
6154	6154-PIERRE PUGET BRETEUIL	PIERRE PUGET BRETEUIL
1002	1002-HOTEL DE VILLE	HOTEL DE VILLE
1224	1224-PLAINE - BIBLIOTHEQUE	PLAINE BIBLIOTHEQUE
1287	1287- STALINGRAD - RÉFORMÉS	STALINGRAD RFORMS
500	500 - IT PLAISIR	IT PLAISIR
5220	5220-ROOSEVELT - ORAN	ROOSEVELT ORAN
6160	6160-153 PARADIS	PARADIS

7.Data Modeling

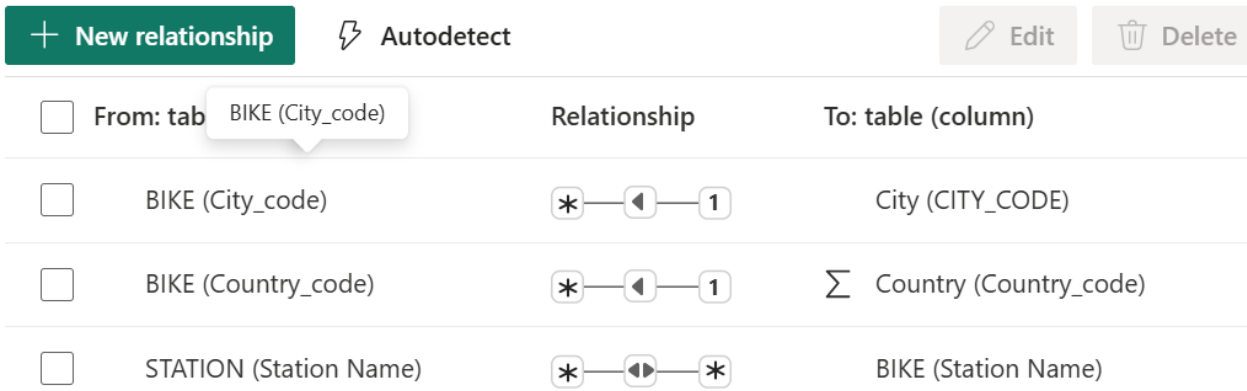
Separated data into :

Fact Table: Bike availability and utilization metrics

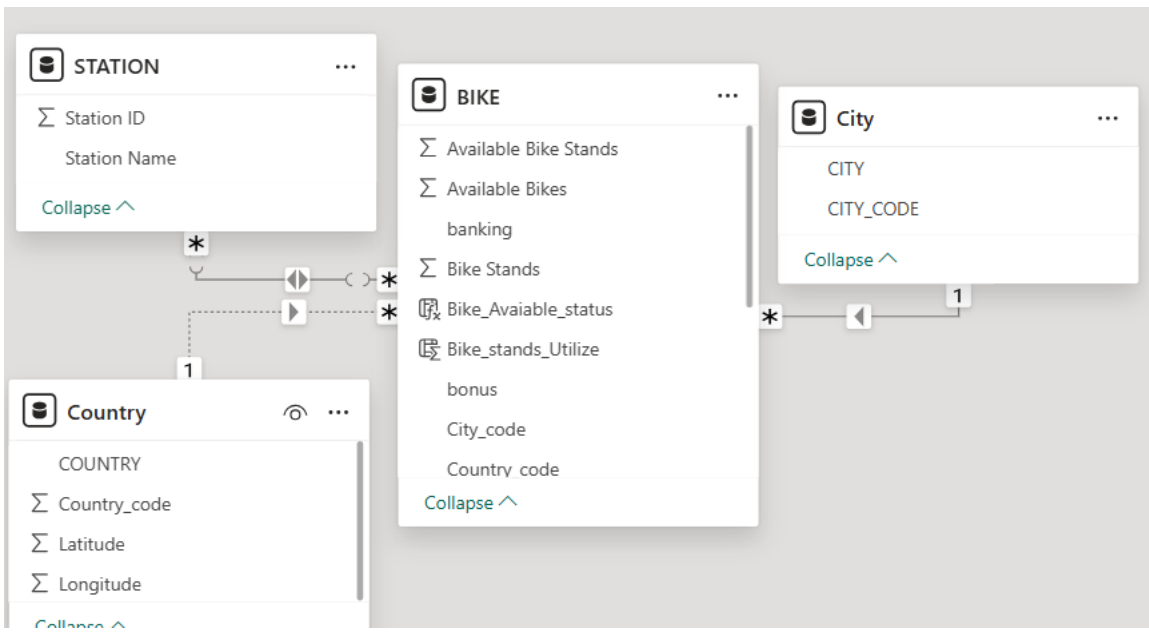
Dimension Tables: Country, City,Station.

- Established relationships using primary and foreign keys.

Manage Relationship



Data Model



DAX & Calculations

Calculated Columns:

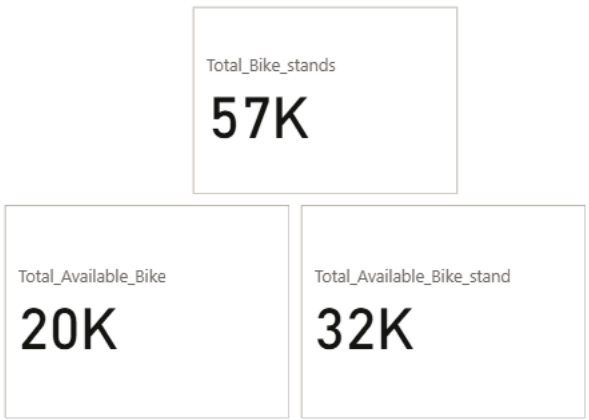
```
Bike_stands_Utilize = BIKE[Bike Stands]-BIKE[Available Bike Stands]
```

```
Bike_Avaiable_status = IF(BIKE[Available Bikes]=0,"No Bike Availabe","Bike Available")
```

DAX MEASURE

- **%No_Bike_Available** = [No_Bike_Available]/[Bike_Count_Status]
- **%Available_Bike** = [Available_Bike_Count]/[Bike_Count_Status]
- **Available_Bike_Count** = CALCULATE(COUNTA(BIKE[Bike_Avaiable_status]),BIKE[Bike_Avaiable_status]="Bike Available")
- **No_Bike_Available** = CALCULATE(COUNTA(BIKE[Bike_Avaiable_status]),BIKE[Bike_Avaiable_status]="No Bike Availabe")
- **Total_Available_Bike** = sum(BIKE[Available Bikes])
- **Total_Available_Bike_stand** = sum(BIKE[Available Bike Stands])
- **Total_Bike_stands** = sum(BIKE[Bike Stands])

DATA VISUALIZTION

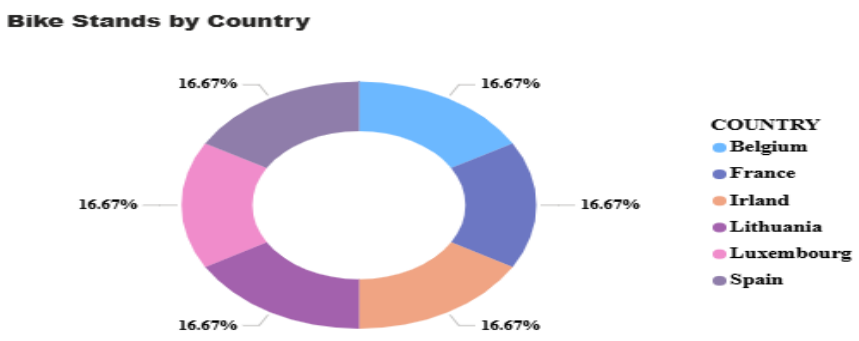


1. Cards

Visuals:

- Total Available Bikes – **20K**
- Total Available Bike Stands – **32K**
- Total Bike Stands – **57K**

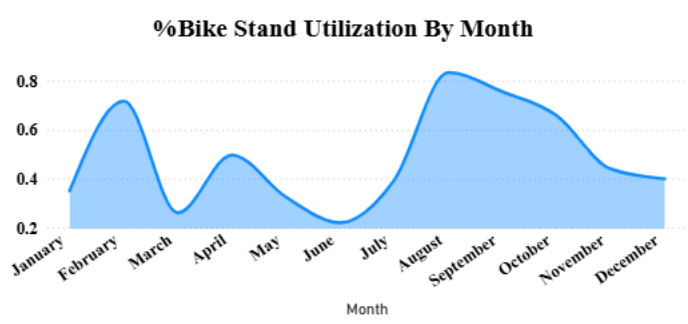
2.Bike Stands by Country (Donut Chart)



Insights:

- Bike stand distribution is **nearly uniform across countries**, reflecting balanced infrastructure investment.
- Performance differences are therefore driven by **city-level demand and operations**, not national infrastructure.
- Supports fair benchmarking across regions.

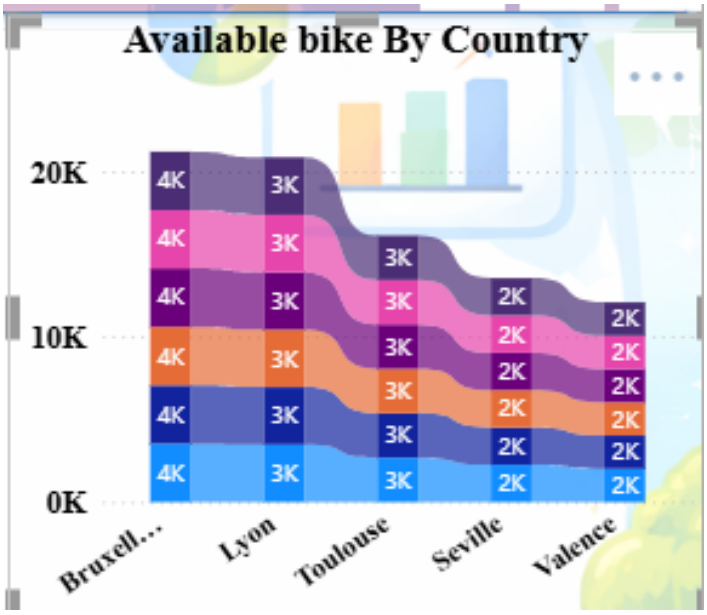
3. % Bike Stand Utilization by Month (Area / Line Chart)



Insights

- **Peak utilization occurs between August and September**, indicating strong seasonal demand.
- **Lowest utilization observed in June**, likely due to weather, holidays, or operational downtime.
- Utilization drops gradually from October to December, suggesting reduced winter demand.

5. Available Bikes by Country



Insight

- **Brussels and Lyon** show the **highest bike availability**, indicating major usage hubs.
- **Seville and Valencia** have comparatively lower availability, possibly due to:
  - Smaller fleet size
  - Higher utilization
  - Limited infrastructure
- Consistent color segments suggest similar station composition across countries.

6. Station Utilization

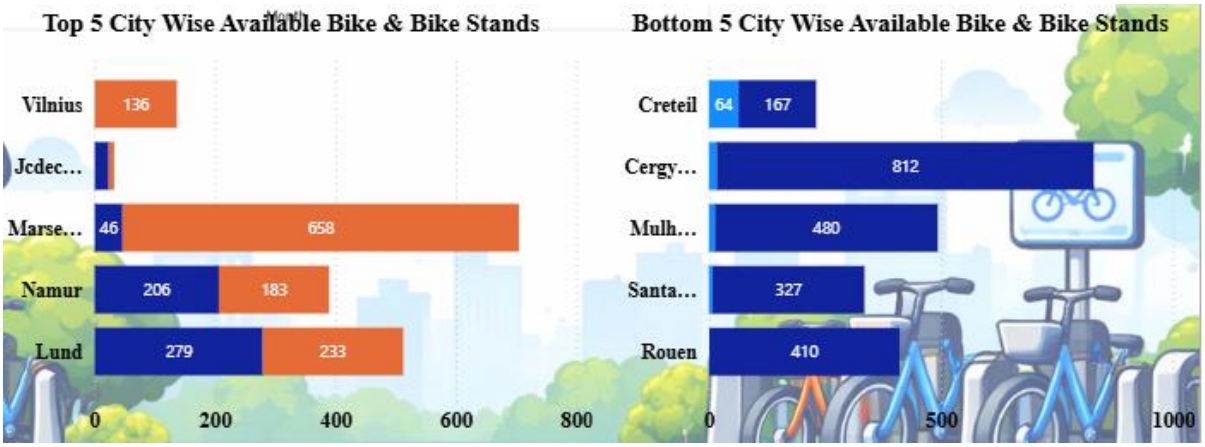
Station	Utilize	%Bike	%I
AOT	20.00%	100.00%	
AOT PRESSENS	47.37%	100.00%	
AVIONS LIBELLULES	20.00%	100.00%	
GLOIRE SAVIGNAC	15.38%	100.00%	
LANGUEDOC NAZARETH	16.67%	100.00%	
MIRAIL UNIVERSIT MTRO	100.00%	100.00%	
NOVEMBRE	97.14%	100.00%	
NOVEMBRE GASTON BERGER	100.00%	100.00%	
NOVEMBRE NGOGOUSSES	66.67%	100.00%	
PERRACHE PETIT	100.00%	100.00%	
SEPTEMBRE CB	40.00%	100.00%	
WAGNER VIVALDI	38.89%	100.00%	
ABBAYE DE LA CAMBRE ABDIJ TER KAMEREN	25.00%	100.00%	
ADEPS	50.00%	100.00%	
ADOLPHE MAX	57.14%	100.00%	
AEROPOLIS	72.00%	100.00%	
AGORA	68.42%	100.00%	
ALBERT THOMAS VILLON	95.00%	100.00%	
ALBERTINE ALBERTINA	20.83%	100.00%	
ALBI AVRANCHES	29.41%	100.00%	
Total	**43.85%**	**75.32%**	

Insights

Several stations show 100% bike availability but low utilization**, indicating demand mismatch.**

- High-performing stations (e.g., >70% utilization) are candidates for:
  - Fleet expansion
  - Priority maintenance
- Low-utilization stations may require:
  - Relocation
  - Promotional strategies
  - Demand analysis
  -

7.Top & Bottom 5 City-wise Available Bike & Bike Stands



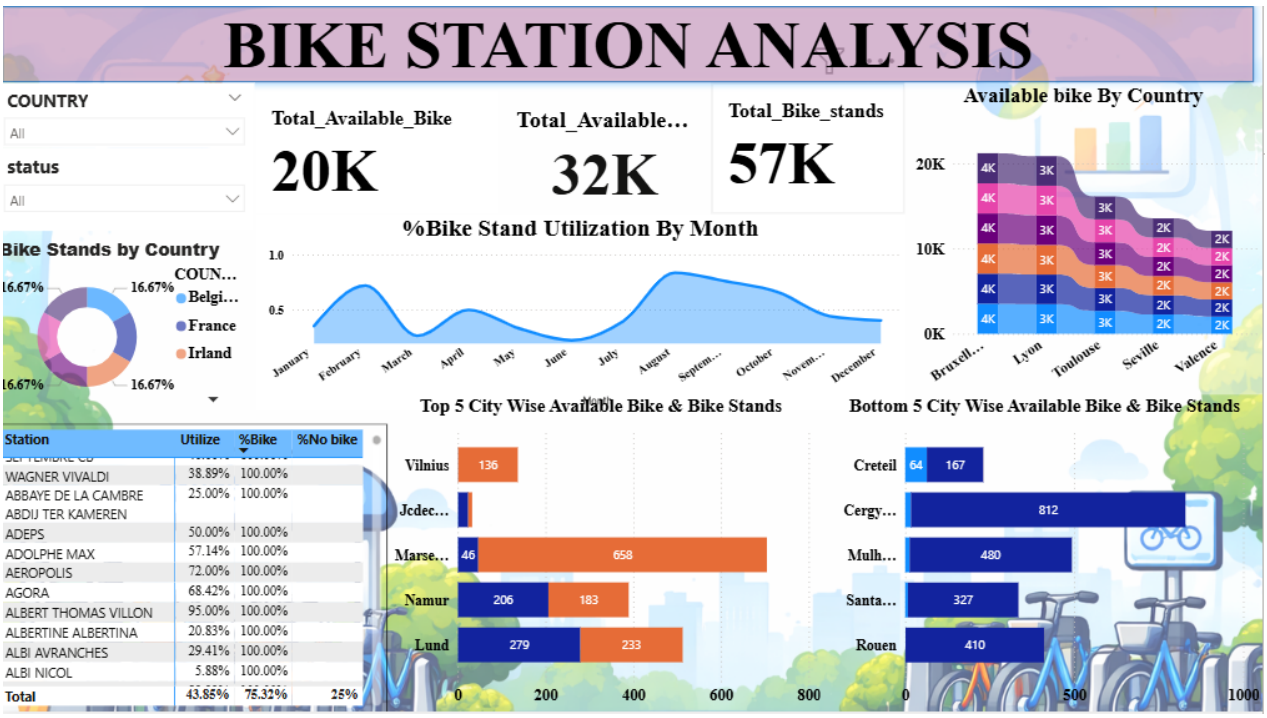
Insights

Top 5 city Insights

- **Marseille** stands out with the **highest bike stand capacity**, but bike availability is comparatively lower → possible high usage.
- **Lund and Namur** show a healthy balance between bikes and stands.
- Cities with high stand count but low bike count may indicate:
  - Strong demand
  - Need for rebalancing or bike redistribution

Bottom 5 city Insights

- **Cergy and Mulhouse** show extremely low bike availability compared to stands.
- Indicates either:
  - Underperforming stations
  - Oversupply of infrastructure
- These cities are strong candidates for:
  - Station consolidation
  - Resource reallocation
  - Pilot programs to boost adoption



ANALYSIS

1. Descriptive Analysis

- The system has **57K total bike stands**, but only **20K bikes** are currently available, indicating a large infrastructure–fleet gap.
- Bike stand distribution across countries is **uniform (~16.67% each)**, showing balanced infrastructure deployment.
- **Brussels and Lyon** have the highest number of available bikes, making them the primary operational hubs.
- **Bike stand utilization peaks in August–September**, while the lowest utilization is observed around **June**.
- Several stations report **100% bike availability but low utilization**, pointing to unused resources.

2. Diagnostic Analysis

- The **mismatch between bike stands and bike availability** suggests inefficient fleet allocation rather than infrastructure shortage.
- Seasonal spikes in late summer likely correlate with:
  - Favorable weather
  - Tourism and commuting demand
- Low utilization in some cities may be due to:
  - Lower population density
  - Poor station placement
  - Limited user awareness
- Cities with high stand capacity but fewer bikes indicate **frequent bike usage or delayed redistribution**.

3. Predictive Analysis

- If current trends continue, **bike demand will increase during Q3 (July–September)** every year.
- Without rebalancing, high-demand cities will face **bike shortages during peak months**, affecting customer satisfaction.
- Low-performing cities will continue to show **underutilized infrastructure**, leading to higher operational costs.
- Winter months (November–January) will likely see **continued decline in utilization**, requiring demand-based scaling.

4. Prescriptive Analysis

- **Implement dynamic bike rebalancing** between high- and low-demand cities to improve utilization.
- Increase bike availability in **Brussels, Lyon, and Marseille** during peak months.
- Optimize or consolidate stations in **low-performing cities** such as Cergy and Mulhouse.
- Use **seasonal forecasting models** to plan maintenance and fleet sizing.
- Introduce **incentives or promotions** in low-utilization areas to stimulate demand.
- Align infrastructure expansion with **actual usage patterns**, not uniform distribution.

Conclusion

The Bike Station Analysis highlights that while the bike-sharing system has a **strong and well-distributed infrastructure**, overall performance is constrained by **uneven utilization across cities and clear seasonal demand patterns**. Peak usage during late summer contrasts sharply with underutilization in certain months and locations, indicating inefficiencies in resource allocation rather than a lack of capacity.

The findings show that **uniform deployment of bike stands does not align with actual demand**, leading to idle assets in low-performing cities and potential shortages in high-demand hubs. Seasonal trends further emphasize the need for proactive planning to maintain service quality during peak periods.

By adopting **data-driven redistribution, seasonal fleet planning, and station-level optimization**, the system can significantly improve utilization, reduce operational waste, and enhance user availability without major infrastructure expansion. Overall, the analysis demonstrates that **strategic optimization, not additional capacity, is the key driver of improved performance and sustainability** in the bike-sharing network.

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